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MULTIVARIABLE CALCULUS • UNIT 7.01

Greens Theorem, Circulation & Planar Flux

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

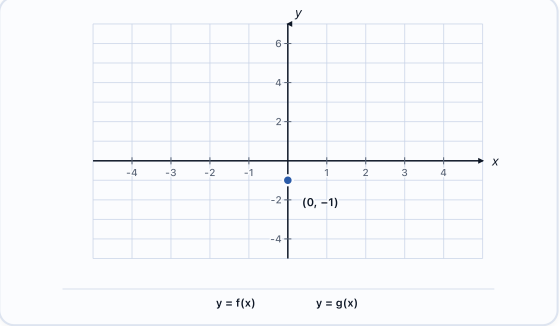
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 C	Q2 B	Q3 C	Q4 A	Q5 A
Q6 B	Q7 D	Q8 A	Q9 C	Q10 C
Q11 B	Q12 D	Q13 D	Q14 C	Q15 B
Q16 B	Q17 A	Q18 A	Q19 A	Q20 B
Q21 A	Q22 B	Q23 B	Q24 D	Q25 B

Greens Theorem, Circulation & Planar Flux Practice Exercises

1 MCQ

Consider the function $z = f(x, y) = x^2 + y^2$. Let $x = 2t$ and $y = 3t$. Use the Multivariable Chain Rule to find $\frac{dz}{dt}$ when $t = 1$.



A 10

B 13

C 26 ✓

D 52

WORKED SOLUTION & VERIFICATION

We are given $z = f(x, y) = x^2 + y^2$, with $x = 2t$ and $y = 3t$. We need to find $\frac{dz}{dt}$ using the Multivariable Chain Rule:
$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$.
First, find the partial derivatives of z with respect to x and y:
$\frac{\partial z}{\partial x} = 2x$
$\frac{\partial z}{\partial y} = 2y$
Next, find the derivatives of x and y with respect to t:
$\frac{dx}{dt} = 2$
$\frac{dy}{dt} = 3$
Now, substitute these into the chain rule formula:
$\frac{dz}{dt} = (2x)(2) + (2y)(3) = 4x + 6y$.
We are asked to find $\frac{dz}{dt}$ when $t = 1$. First, find the values of x and y when $t = 1$:
$x = 2(1) = 2$
$y = 3(1) = 3$
Finally, substitute these values of x and y into the expression for $\frac{dz}{dt}$:
$\left.\frac{dz}{dt}\right|_{t=1} = 4(2) + 6(3) = 8 + 18 = 26.$

2 MCQ

Evaluate the double integral of the function $f(x, y) = 2xy$ over the rectangular region $R = [0, 2] \times [0, 3]$ using Fubini's Theorem. Set up the iterated integral by integrating with respect to y first, then x.

A 12	**B 18** ✓
C 6	D 24

WORKED SOLUTION & VERIFICATION

To evaluate the double integral $\iint_R 2xy\, dA$ where $R = [0, 2] \times [0, 3]$, we can use Fubini's Theorem. We will set up the iterated integral by integrating with respect to y first, then x:

$$\int_0^2 \int_0^3 2xy\, dy\, dx$$

First, integrate with respect to y, treating x as a constant:

$$\int_0^3 2xy\, dy = [xy^2]_{y=0}^{y=3} = x(3^2) - x(0^2) = 9x$$

Now, integrate the result with respect to x:

$$\int_0^2 9x\, dx = [\frac{9x^2}{2}]_{x=0}^{x=2} = \frac{9(2^2)}{2} - \frac{9(0^2)}{2} = \frac{9(4)}{2} = \frac{36}{2} = 18$$

Therefore, the value of the double integral is 18.

3 MCQ

For $f(x, y) = 7x^2 + 4y^2$, find $f_y(-3, 1)$.

A -24	B 4
C 8 ✓	D -42

WORKED SOLUTION & VERIFICATION

Applying Partial Derivatives principles, $f_y = 8y$, so $f_y(-3, 1) = 8(1) = 8$. Therefore, the verified correct option is 8.

4 MCQ

Consider the function $f(x, y) = x^2 + y^2$. Find the value of $f_{xx}(2, 3)$.

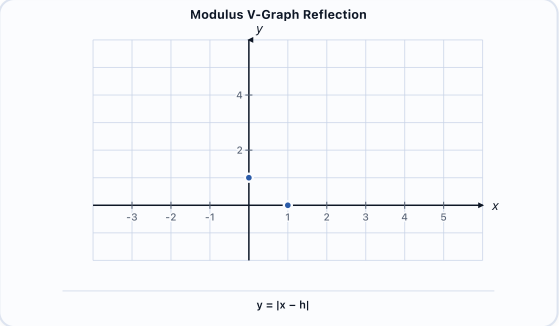
A 2 ✓	B 4
C 6	D 0

WORKED SOLUTION & VERIFICATION

First, we find the first partial derivative of $f(x, y)$ with respect to x: $f_x(x, y) = \frac{\partial}{\partial x}(x^2 + y^2) = 2x$.
Next, we find the second partial derivative of $f(x, y)$ with respect to x (which is f_{xx}): $f_{xx}(x, y) = \frac{\partial}{\partial x}(f_x(x, y)) = \frac{\partial}{\partial x}(2x) = 2$.
Finally, we evaluate $f_{xx}(x, y)$ at the point $(2, 3)$. Since $f_{xx}(x, y)$ is a constant, its value at any point is 2. Therefore, $f_{xx}(2, 3) = 2$.

5 MCQ

Let the linear scalar potential be $h(u, y) = 15u + 8y - 16$. Compute the squared gradient norm $\|\nabla h\|^2$.



- **A** 289 ✓
- **B** 293
- **C** 281
- **D** 304

WORKED SOLUTION & VERIFICATION

Since $\nabla h = \langle 15, 8 \rangle$, $\|\nabla h\|^2 = 15^2 + 8^2 = 289$.

6 MCQ

Consider the surface S which is the portion of the plane $z = 2x + 3y + 1$ lying above the rectangular region $R = [0, 1] \times [0, 2]$ in the xy-plane. Calculate the surface area of S.

- **A** $\sqrt{14}$
- **B** $2\sqrt{14}$ ✓
- **C** $3\sqrt{14}$
- **D** $4\sqrt{14}$

WORKED SOLUTION & VERIFICATION

The surface area of a surface $z = g(x, y)$ above a region R is given by the integral $\iint_R \sqrt{1 + (g_x)^2 + (g_y)^2}\, dA$. Here, $g(x, y) = 2x + 3y + 1$. The partial derivatives are $g_x = 2$ and $g_y = 3$. Therefore, $\sqrt{1 + (g_x)^2 + (g_y)^2} = \sqrt{1 + 2^2 + 3^2} = \sqrt{1 + 4 + 9} = \sqrt{14}$. The integral becomes $\iint_R \sqrt{14}\, dA$, where $R = [0, 1] \times [0, 2]$. This evaluates to $\sqrt{14} \times \text{Area}(R)$. The area of the rectangle R is $(1 - 0) \times (2 - 0) = 2$. So, the surface area is $2\sqrt{14}$.

7 MCQ

Consider the vector field $\mathbf{u}(s, y, z) = \langle 22s,\ 7y,\ 7z \rangle$ on R^3. Compute the scalar divergence $\nabla \cdot \mathbf{u}$.

- **A** 58
- **B** 40
- **C** 29
- **D** 36 ✓

WORKED SOLUTION & VERIFICATION

The divergence is $22 + 7 + 7 = 36$.

8 MCQ

Consider the vector field $\mathbf{F}(x, y) = \langle 3x, 2y \rangle$. Evaluate the vector field at the point $(2, 4)$.

- **A** $\langle 6, 8 \rangle$ ✓
- **B** $\langle 8, 6 \rangle$
- **C** $\langle 5, 6 \rangle$
- **D** $\langle 6, 5 \rangle$

WORKED SOLUTION & VERIFICATION

The vector field is given by $\mathbf{F}(x, y) = \langle 3x, 2y \rangle$. To evaluate the vector field at the point $(2, 4)$, we substitute $x = 2$ and $y = 4$ into the components of the vector field.
First component: $3x = 3(2) = 6$.
Second component: $2y = 2(4) = 8$.
Therefore, $\mathbf{F}(2, 4) = \langle 6, 8 \rangle$.

9 MCQ

Use Stokes' theorem to find the circulation $\oint_C \mathbf{F} \cdot d\mathbf{r}$ of $\mathbf{F} = \langle -y, x, z \rangle$ around the circle $x^2 + y^2 = 9$ in the plane $z = 0$, oriented counterclockwise seen from above.

- **A** 36π
- **B** -18π
- **C** 18π ✓
- **D** 9π

WORKED SOLUTION & VERIFICATION

$\nabla \times \mathbf{F} = \langle 0, 0, 2 \rangle$. Taking the flat surface inside C with $\mathbf{n} = \mathbf{k}$, $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{k}\, dS = 18\pi$.

10 MCQ

Use Stokes' theorem to find the circulation $\oint_C \mathbf{F} \cdot d\mathbf{r}$ of $\mathbf{F} = \langle y, -x, xz \rangle$ around the boundary of the square $[0, 2] \times [0, 2]$ in the plane $z = 0$, oriented counterclockwise seen from above.

- **A** 8
- **B** -4
- **C** -8 ✓
- **D** -16

WORKED SOLUTION & VERIFICATION

$\nabla \times \mathbf{F} = \langle 0, -z, -2 \rangle$. Taking the flat surface inside C with $\mathbf{n} = \mathbf{k}$, $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{k}\, dS = -8$.

11 MCQ

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x, y, z \rangle$ across the sphere $x^2 + y^2 + z^2 = 4$.

- **A** 64π
- **B** 32π ✓
- **C** 16π
- **D** -32π

WORKED SOLUTION & VERIFICATION

$\nabla \cdot \mathbf{F} = 3$. The flux equals $\iiint_E \nabla \cdot \mathbf{F}\, dV = 32\pi$.

12 MCQ

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x^3, y^3, z^3 \rangle$ across the cube $[0, 2]^3$.

A 192	B 48
C -96	D 96 ✓

WORKED SOLUTION & VERIFICATION

$\nabla \cdot \mathbf{F} = 3x^2 + 3y^2 + 3z^2$. The flux equals $\iiint_E \nabla \cdot \mathbf{F}\, dV = 96$.

13 MCQ

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle xz, 0, z^2 \rangle$ across the cube $[0, 1]^3$.

A $\frac{3}{4}$	B $-\frac{3}{2}$
C 3	D $\frac{3}{2}$ ✓

WORKED SOLUTION & VERIFICATION

$\nabla \cdot \mathbf{F} = 3z$. The flux equals $\iiint_E \nabla \cdot \mathbf{F}\, dV = \frac{3}{2}$.

14 MCQ

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x^2, y^2, z^2 \rangle$ across the closed cylinder $x^2 + y^2 \leq 1,\ 0 \leq z \leq 3$.

A 18π	B $\frac{9\pi}{2}$
C 9π ✓	D -9π

WORKED SOLUTION & VERIFICATION

$\nabla \cdot \mathbf{F} = 2x + 2y + 2z$. The flux equals $\iiint_E \nabla \cdot \mathbf{F}\, dV = 9\pi$.

15 MCQ

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x^3, y^3, z^3 \rangle$ across the sphere $x^2 + y^2 + z^2 = 4$.

A $-\frac{384\pi}{5}$	B $\frac{384\pi}{5}$ ✓
C $\frac{768\pi}{5}$	D $\frac{192\pi}{5}$

WORKED SOLUTION & VERIFICATION

$\nabla \cdot \mathbf{F} = 3x^2 + 3y^2 + 3z^2$. The flux equals $\iiint_E \nabla \cdot \mathbf{F}\, dV = \frac{384\pi}{5}$.

16 MCQ

Find the flux $\iint_S \mathbf{F} \cdot d\mathbf{S}$ of $\mathbf{F} = \langle y, x, 3 \rangle$ across the disk $x^2 + y^2 \leq 1,\ z = 0$, with upward normal.

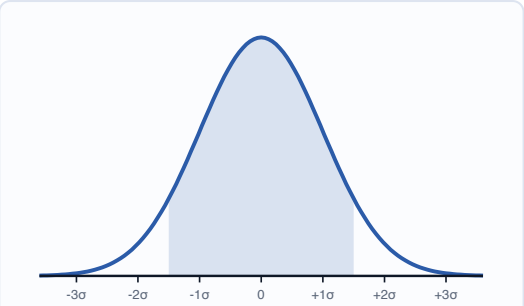
A $\frac{3\pi}{2}$	B 3π ✓
C 6π	D -3π

WORKED SOLUTION & VERIFICATION

On the disk $\mathbf{n} = \mathbf{k}$, so the flux is $\iint_D F_3\, dA = 3\pi$.

17 MCQ

Find the flux $\iint_S \mathbf{F} \cdot d\mathbf{S}$ of $\mathbf{F} = \langle 0, 0, z \rangle$ across the part of the plane $x + y + z = 1$ in the first octant, with upward normal.



- **A** $\frac{1}{6}$ ✓
- **B** $\frac{1}{12}$
- **C** $\frac{1}{3}$
- **D** $-\frac{1}{6}$

WORKED SOLUTION & VERIFICATION

Write $z = -x - y + 1$; $d\mathbf{S} = \langle 1, 1, 1 \rangle \, dA$. The flux is $\iint_D \mathbf{F} \cdot \langle 1, 1, 1 \rangle \, dA = \frac{1}{6}$.

18 MCQ

Calculate the flux $\iint_S \mathbf{F} \cdot d\mathbf{S}$ of $\mathbf{F} = \langle 0, 0, z \rangle$ across the part of the plane $x + y + z = 3$ in the first octant, with upward normal.

- **A** $\frac{9}{2}$ ✓
- **B** 9
- **C** $-\frac{9}{2}$
- **D** $\frac{9}{4}$

WORKED SOLUTION & VERIFICATION

Write $z = -x - y + 3$; $d\mathbf{S} = \langle 1, 1, 1 \rangle \, dA$. The flux is $\iint_D \mathbf{F} \cdot \langle 1, 1, 1 \rangle \, dA = \frac{9}{2}$.

19 MCQ

Find the flux $\iint_S \mathbf{F} \cdot d\mathbf{S}$ of $\mathbf{F} = \langle 1, 1, 1 \rangle$ across the part of the plane $x + y + z = 3$ in the first octant, with upward normal.

- **A** $\frac{27}{2}$ ✓
- **B** 27
- **C** $-\frac{27}{2}$
- **D** $\frac{27}{4}$

WORKED SOLUTION & VERIFICATION

Write $z = -x - y + 3$; $d\mathbf{S} = \langle 1, 1, 1 \rangle \, dA$. The flux is $\iint_D \mathbf{F} \cdot \langle 1, 1, 1 \rangle \, dA = \frac{27}{2}$.

20 MCQ

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle 2x, -y, 3z \rangle$ across the closed cylinder $x^2 + y^2 \leq 4,\ 0 \leq z \leq 2$.

A 64π

B 32π ✓

C 16π

D -32π

WORKED SOLUTION & VERIFICATION

$\nabla \cdot \mathbf{F} = 4$. The flux equals $\iiint_E \nabla \cdot \mathbf{F}\, dV = 32\pi$.

21 MCQ

Consider the vector field $\mathbf{F}(x, y) = \langle -y, x \rangle$. Evaluate the line integral $\oint_C \mathbf{F} \cdot d\mathbf{r}$ where C is the circle $x^2 + y^2 = 4$ oriented counterclockwise, using Green's Theorem.

A 8π ✓

B 0

C 4π

D 16π

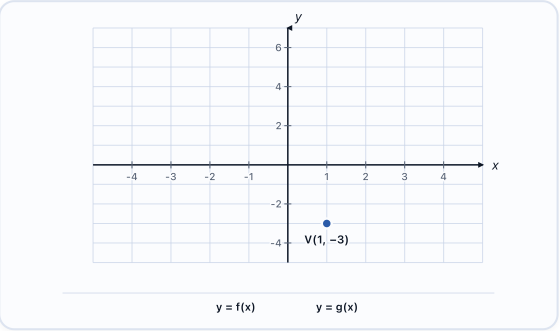
WORKED SOLUTION & VERIFICATION

By Green's Theorem, for $\mathbf{F}(x, y) = \langle P, Q \rangle = \langle -y, x \rangle$ over the disk $D : x^2 + y^2 \leq 4$ of radius $r = 2$:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA = \iint_D (1 - (-1))\, dA = 2 \cdot \text{Area}(D) = 2(\pi \cdot 2^2) = 8\pi.$$

22 MCQ

Consider the vector field $\mathbf{F}(x, y) = \langle x^2, y \rangle$. According to Green's Theorem, the line integral of $\mathbf{F}$ around a simple closed curve C is equal to the double integral of $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$ over the region D enclosed by C, where $\mathbf{F} = \langle P, Q \rangle$. For the given vector field $\mathbf{F}(x, y) = \langle x^2, y \rangle$, identify the value of $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$.



A x^2

B 0 ✓

C y

D -2x

WORKED SOLUTION & VERIFICATION

The given vector field is $\mathbf{F}(x, y) = \langle P(x, y), Q(x, y) \rangle = \langle x^2, y \rangle$.
According to Green's Theorem, we need to compute $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$.
First, find the partial derivative of Q with respect to x: $\frac{\partial Q}{\partial x} = \frac{\partial}{\partial x}(y) = 0$.
Next, find the partial derivative of P with respect to y: $\frac{\partial P}{\partial y} = \frac{\partial}{\partial y}(x^2) = 0$.
Therefore, $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0 - 0 = 0$.

23 MCQ

Calculate $\nabla f(-2, 2)$ for $f(x, y) = 2x^2 + 7y^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	2	7	9

A (28, -8)

B **(-8, 28)** ✓

C (-4, 14)

D (8, -28)

WORKED SOLUTION & VERIFICATION

$\nabla f = (4x, 14y)$, so $\nabla f(-2, 2) = (-8, 28)$.

24 MCQ

Obtain $\nabla f(-5, -3)$ for $f(x, y) = 2x^2 + 3y^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	2	3	5

A (-10, -9)

B (20, 18)

C (-18, -20)

D **(-20, -18)** ✓

WORKED SOLUTION & VERIFICATION

$\nabla f = (4x, 6y)$, so $\nabla f(-5, -3) = (-20, -18)$.

25 MCQ

Calculate $\nabla f(1, -6)$ for $f(x, y) = x^2 + 6y^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	1	6	7

A (-72, 2)

B **(2, -72)** ✓

C (1, -36)

D (-2, 72)

WORKED SOLUTION & VERIFICATION

$\nabla f = (2x, 12y)$, so $\nabla f(1, -6) = (2, -72)$.